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Project Title:	GRADIENT DRIVEN FLUCTUATIONS		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
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Human Research Program Elements:	None		
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Zip Code:	93106-9530	**Congressional District:**	23
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Task Description:	This work is part of an international collaboration between my own group at UC Santa Barbara funded by NASA, and that of professor Marzio Giglio at the University of Milan, funded by ESA/ASI. The flight hardware will be provided by ESA. Our aim is to carry out experimental studies of the fluctuations in liquids and liquid mixtures subjected to gradients. It is well known that macroscopic concentration and/or temperature disturbances relax to a homogeneous state via diffusion. Diffusion is caused by random molecular motion, and a common belief is that the fluctuations occur only on molecular length scales. This picture is wrong, and it is now understood that diffusion is associated with large amplitude, long wavelength fluctuations that restlessly accompany the decay to equilibrium. The fluctuations involved in a diffusion process are large, both in amplitude and spatial extent, and with the support from our first NASA grant, we were able to visualize them using the shadowgraph method. In order to provide a quantitative characterization of the fluctuations, we extended the theory of the shadowgraph, so as to provide quantitative results for the spatial power spectrum of the fluctuations, S(q). This technique allows us to measure the mean squared amplitude of the fluctuations as a function of wave vector q, i.e. S(q) . We have found that the shadowgraph method excels at very small wave vectors. The amplitude of the fluctuations diverges as q^-4 for intermediate wave vectors, so much of the interest will be at very small q, and we have thus focused our attention on the shadowgraph method. The divergence originates from the coupling of what are basically normal equilibrium velocity fluctuations with a macroscopic concentration and/or temperature gradient. Below a cutoff wave vector, the amplitude of the fluctuations is quenched to a constant value in the presence of gravity, but is predicted to continue diverging when gravity is removed. This divergence is predicted to be limited only by the finite size of the sample. Two of us (A. Vailati and M. Giglio) have reported on the observation of giant fluctuations during a free diffusion process occurring in ordinary fluids. These results provide strong evidence that giant concentration fluctuations are a genuinely universal phenomenon associated with diffusion caused by a gradient. The broad objectives of this work are to achieve a better understanding of the manner in which temperature and/or concentration gradients in fluid mixtures enhance spontaneous fluctuations. We also seek to elucidate the surprising role that gravity plays in suppressing such gradient driven fluctuations. Current predicitions are that removing gravity will result in extremely large long-wavelength fluctuations, limited in amplitude only by the size of the device containing the fluid. Further general objectives are to study the time dependence of such fluctuations as a function of the fluctuation wavelength. We intend to carry out such studies for both single-component fluids and for fluid mixtures. This experimental program impacts the way we understand diffusion processes and how we should model them when a temperature gradient or concentration gradient is present. To give one of many possible examples, when a protein crystal grows, it depletes the nutrient from the solute in which it is immersed. This depletion produces a concentration gradient, which in turn produces giant nonequilibrium concentration fluctuations. Thus, the current studies may impact our understanding of the growth process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The main benefits are twofold. First, we have already developed new optical instrumentation making the equivalent of extremely low-angle light scattering possible, and we are gaining fundamental knowledge concerning the behavior of fluids and mixtures subjected to gradients. Our data will be used to test basic theories of fluid mechanics for non-equilibrium fluids.
	Final Summary for the Entire Grant Period: NASA Grant NNC04GA45G During the 5 year period for which research funding was received, the problem of spontaneous fluctuations in fluids that are not in equilibrium was investigated experimentally. The work was carried out as a joint program involving collaboration between the UC Santa Barbara group and that of Professor Marzio Giglio at the University of Milan. This work resulted in a European Space Agency flight project called Gradflex, which flew for 11 days aboard the Foton-M3 mission, September 14-26, 2007. One tends naively to think of a laboratory-sized sample of a fluid, such as water or air, as having the same density, pressure and temperature throughout its volume. This view is actually quite accurate, but incomplete. Although each property, such as temperature has a very well defined value throughout the sample, there are deviations from the average at each point in the sample, and these deviations, which are called fluctuations, vary with time at any given point as well. Understanding such fluctuations has proved extremely useful, both for research purposes and in practical terms as well. The technique of static light scattering can be used to measure the amplitude of fluctuations in any property, e.g. density, upon which the refractive index depends. Not only can the amplitude be measured, the exact manner in which the fluctuations come and go in time can be measured using dynamic light scattering. Such measurements have been used for many years to investigate the fundamental thermodynamic (e.g. compressibility, specific heat ratio, molecular weights) and transport (viscosity, diffusivity, thermal diffusivity) properties of fluids. By measuring the scattered intensity as a function of scattering angle it is also possible to measure the actual spatial extent of fluctuations, provided this spatial scale lies in the range from about 0.03 micro-meters to a few micro-meters. Except near a critical point, the range (spatial size) of the fluctuations is below the lower end of the measurement capability for a fluid or fluid mixture in thermal equilibrium. For systems such as polymer solutions, systems undergoing gelation, aggregation, or crystallization, the range of the fluctuations often become so large as to be impossible to measure. Another situation in which the range of fluctuations becomes very large is when a fluid is held in a non-equilibrium state, as for example by passing heat through the sample. This situation is extremely interesting from a basic science point of view, because it is one of the simplest possible cases of a non-equilibrium system, and such systems are not well understood at present. The goal of this project has been to provide accurate experimental data for the properties of the fluctuations in a thin fluid layer of a single-component fluid with one surface (the upper one on Earth) held at a higher temperature than the other one. There are rather detailed theoretical predictions of such fluctuations, and they are predicted to be very dependent on whether gravity is present on not. In particular theories predict that both the range and amplitude of the fluctuations should increase greatly upon removing gravity. The same theories also predict that the exact manner in which fluctuations build up and die down is sensitively dependent on gravity. On earth fluctuations are predicted to actually propagate like waves, while no such propagation is predicted to occur in the absence of gravity.

Task Progress:	To measure the very long-range fluctuations predicted by theory requires choice of the proper technique. The only reasonable options were judged to be ultra-low angle laser light scattering or the recently emerging method of the quantitative shadowgraph, which was invented by the author in collaboration with Mingming Wu, Steven Trainoff and Guenter Ahlers. The later method has the advantages that it is very rugged, is self-calibrating to a reasonable degree, and can provide both static and dynamic information regarding the fluctuations. Its major drawback is that the method is new, and is still undergoing development, both with regard to the actual instrumentation and in our theoretical understanding of how to interpret the data. It was determined early in the project that the only viable possibility was the shadowgraph, however. In this method, a plane parallel beam of light is passed through the sample and is then allowed to fall on a sensor, such as a digital camera, some distance from the sample. Fluctuations in the refractive index of the sample result in fluctuations in the local intensity of the beam. By taking a sequence of images and dividing one into another (or by subtracting one from another), the small (less than one percent) fluctuations in intensity are revealed. These ratio or difference images can then be analyzed to extract the same information that would be obtained by having a light scattering instrument with as half as many detectors as the sensor has pixels. This large number of effective detectors is a huge advantage for this method, as it permits a great deal of averaging. By spacing the images correctly in time, it is possible to extract both the static and dynamic information that would normally be obtained using conventional static and dynamic light scattering. A suitable ground-based prototype shadowgraph instrument was developed in house at UC Santa Barbara, using support from NASA. This prototype performed well and provided excellent data, which has now been published in the prestigious journal, "Physical Review Letters". This data was in very good, but certainly not perfect agreement with existing theory. The question of whether the theory needs more work could not be answered unequivocally because the upper surface used to confine the sample was not an excellent thermal conductor, for reasons of cost. In large part, it was the existence of this prototype that convinced the European Space Agency of the viability of the proposed flight experiment. A major effort was undertaken to assist contractors with the development of suitable flight hardware; an effort that I do not believe was 100% successful, however. The resulting instrument should have been superior to the prototype both optically and in terms of thermal performance. Failure to utilize the highest optical grade of sapphire (admittedly expensive) for the optical window above the sample resulted in considerable beam degradation, and it remains unclear as to exactly how, or even whether, the effects of such degradation can be fully incorporated into the data analysis. At a minimum, this problem has made data analysis for the flight instrument extremely challenging! Using the admittedly imperfect flight hardware, a vast number of images have been obtained for the flight sample, which consists of a 3 mm thick layer of carbon disulphide (chosen solely for its relatively large signal). Images were obtained with the two surfaces held at temperatures differing by values ranging from 0 to 30 degrees Centigrade, both on Earth and in space. Preliminary analysis of the flight data relative to the ground-based data shows clearly that the fluctuations are indeed greatly enhanced by removing gravity, and in addition their dynamics do correspond to propagating wave-like behavior on Earth and not in space. Thus the data confirm the theoretical predictions in a qualitative sense. However, the goal of providing an accurate quantitative test of the theory was not achieved during the initial grant period, and in fact this goal remains elusive, in large measure because of the optical problems with the flight hardware. What has been achieved however represents a considerable step forward. For the first time, a method has been developed that allows measurements to be made for the very small fluctuations caused by heating the upper surface of a fluid sample. To within very reasonable accuracy, such measurements carried out using the prototype show that the theory is on the right track. The vast amount of data obtained using the flight instrument remains under analysis. I think it will be capable of testing the theory rather well for ground-based situations, but I am not so sanguine about the flight data. In addition, one must count as a success, the continued development of the quantitative shadowgraph method as a technique equivalent to static and dynamic light scattering, but which extends light scattering capability to a previously inaccessible regime, namely that of systems with fluctuations on length scales in excess of a few microns. If fact, we have now demonstrated that the method can be used to measure fluctuations with spatial scales as long as thousands of micron, as opposed to a few microns, which is the practical limit for conventional light scattering. This method is now being taken up by both ESA and NASA jointly for the study of colloidal systems.
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